Data Visualization with R

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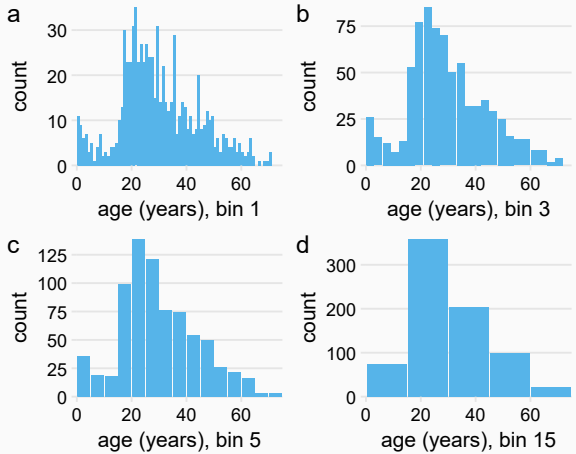
Visualizing Distributions

Distributions

- How a particular variable is distributed in a dataset
- Statistics: Mean, median, mode, standard deviation, skewness, kurtosis
- Visualizations: Histogram, density plot, boxplot, violin plot

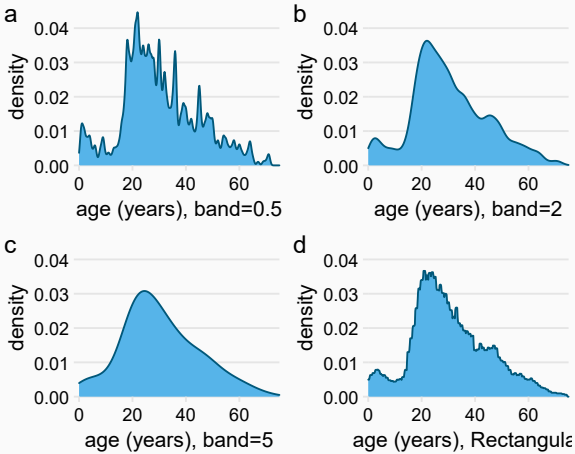
Histograms

- Filled rectangles
- Bin = Width of rectangle = range of values in that bin
- Height of rectangle = number of observations in that bin
- Bin width matters for storytelling



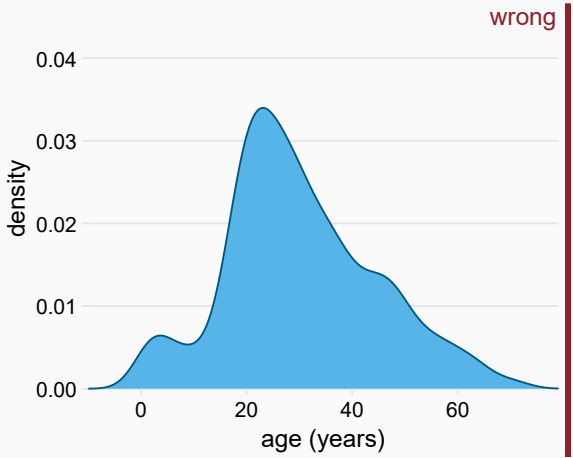
Density plots

- Curve estimated from the data - smooth version of a histogram
- Area under the curve = 1
- Height of curve = density of observations at that value
- Density interpretation: Probability of observing a value in a given range = area under the curve for that range
- Bandwidth matters for storytelling



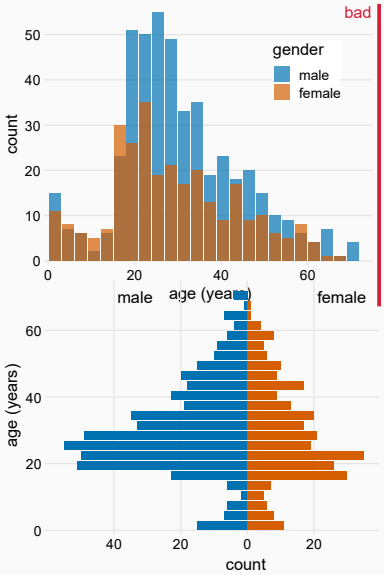
Density plots

- Pitfall: smooth across gaps in data or where there are no observations.



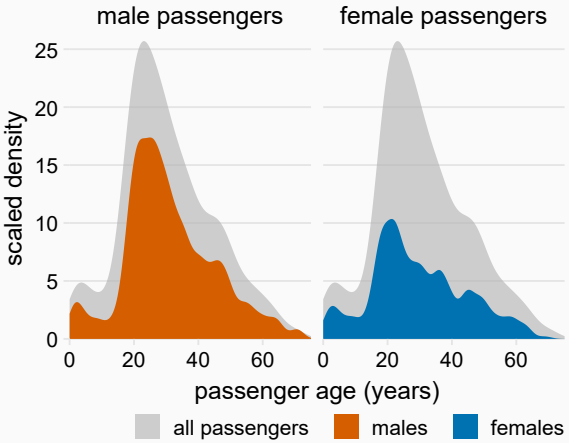
Two or more distributions

- Avoid stacked histograms - hard to compare values
- Bars start at different points
- Use side-by-side histograms or density plots instead



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Two or more distributions

- Density plots extend nicely to more than one distribution
- Use color to separate distributions
- Transparency can help with overlapping distributions

